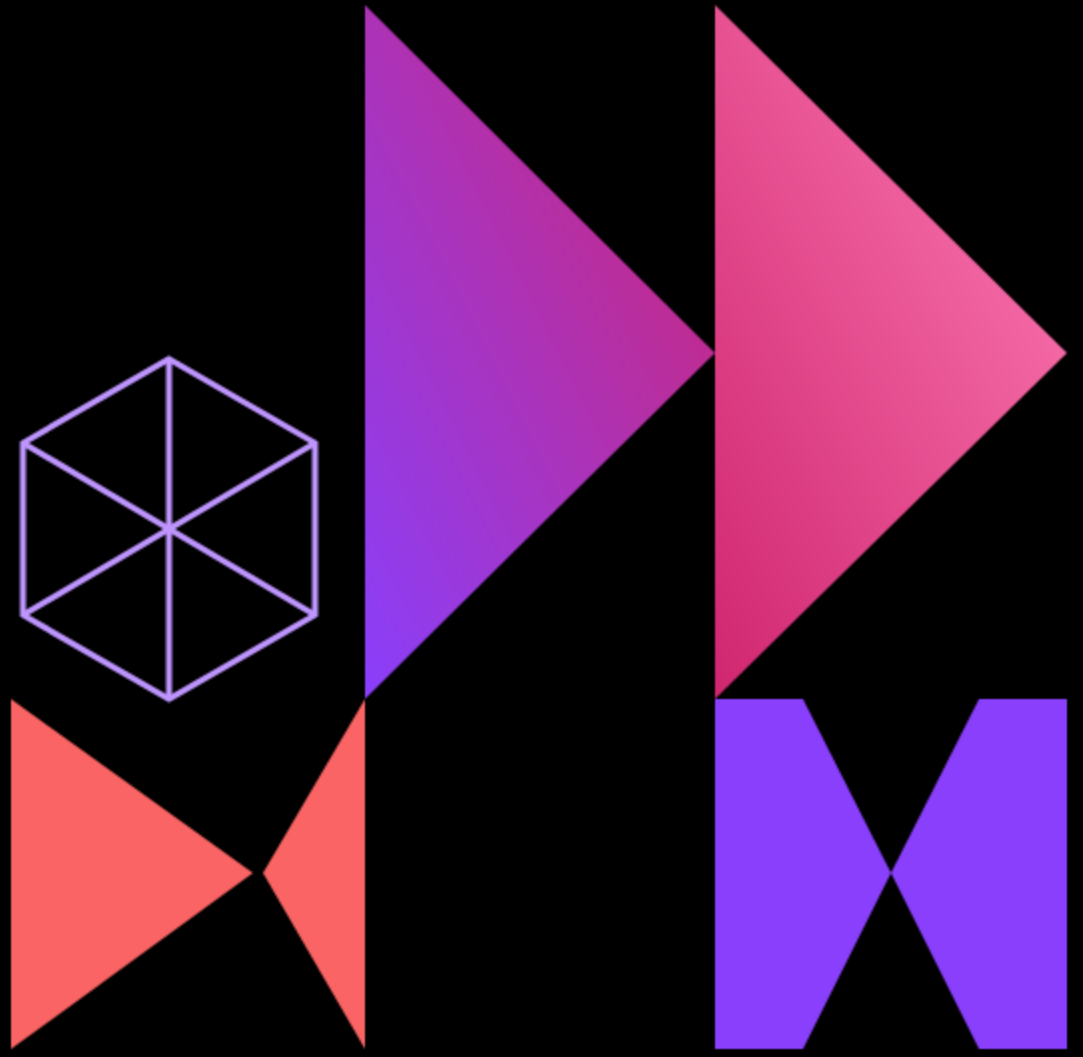
# Instructor-led
# Lab session (1203): Introduction

**Build a Management Report Agent on watsonx**
**Orchestrate and embed it in Planning Analytics**



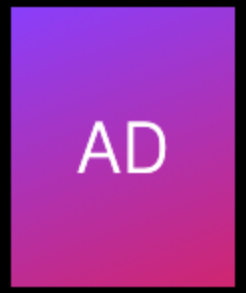
## Access Code: techxLAB-1203-16f7

**Aisha Mohammed Farooq Darga**
AI Engineer, Service Engineering - US Select Territory

**Sami El Cheikh**
Senior Product Manager, IBM Planning Analytics

IBM **TechXchange**

## Instructor-led Lab Guide Team


**Aisha Mohammed Farooq Darga**

AI Engineer, Service Engineering - US Select Territory

–

aisdarg1@in.ibm.com




**Sami El Cheikh**

Senior Product Manager, IBM Planning Analytics

–

sami.el.cheikh@ibm.com

WHAT YOU WILL BUILD

# A management report agent, inside Planning Analytics

By the end, an agent in your PA workspace builds a PDF management report on request.

01

Reads live data from the 24Retail database through the PA MCP server

02

Produces Budget vs Actual tables, outlier checks and variance drivers

03

Builds a PDF report with a custom Python tool

04

Stores it in IBM Cloud Object Storage and returns a 1-hour link

05

Runs inside the PA workspace chat panel

### Agent

An LLM with instructions and a short list of tools. You import it from YAML.

### Toolkit

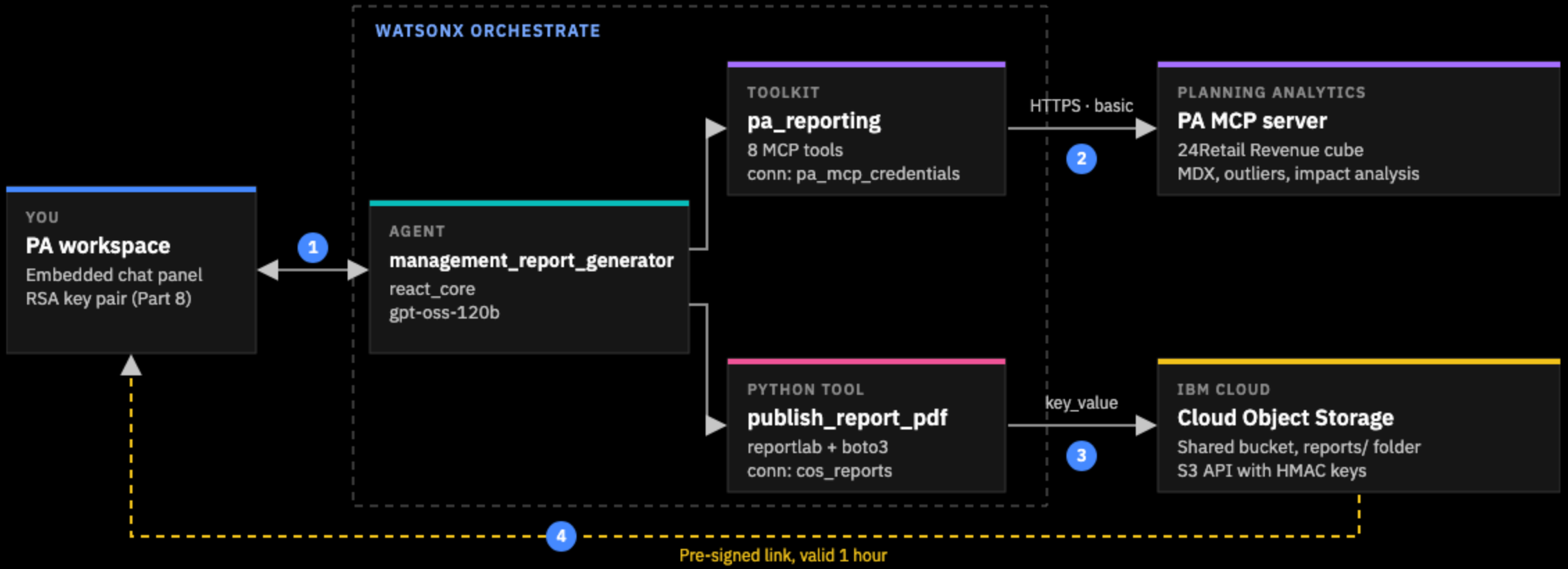
A bundle of tools published by an MCP server. Here: 8 scoped PA tools.

### Connection

Holds credentials. `basic` for PA, `key_value` for COS.

ARCHITECTURE

# How a question becomes a PDF




1. Ask for a report in the PA chat panel
2. Agent reads 24Retail data through the MCP toolkit
3. Agent passes the tables to your Python tool, which uploads the PDF
4. The link comes back to the chat

PLAN

# Your 75 minutes

The instructors provide the agent YAML and the tool file. Clean-up (Part 10) happens after the session.




| BLOCK | MIN | WHAT YOU DO | GUIDE |
|---|---|---|---|
| **Check** | **3** | Pre-work done. Run orchestrate --version and connections list. | Parts 0–1 |
| **Connect PA** | **10** | Create the Basic connection and register the pa_reporting toolkit. | Part 2 |
| **Agent** | **13** | Import the provided agent YAML. Test the executive summary and regional prompts. | Parts 3–4 |
| **PDF tool** | **25** | COS connection, import the provided tool, make 3 YAML edits, run prompts 1, 2 and 7. | Parts 5–7 |
| **Embed** | **12** | Deploy, enter config and key in PA Integrations, test in the chat panel. | Part 8 |
| **Buffer** | **12** | Token expiries, typos, 401 errors. | |

BEFORE YOU START

# Rules, tips and what you need

## Security rules

- **Never paste keys** into chat, tickets or screenshots
- Type secrets with `read -s`, then `unset`
- Keep secrets out of the project folder
- Use lab-only credentials and delete them afterwards

## Tips

- Test in the wxO **preview panel**, not `orchestrate chat ask`
- Use **member IDs** (100, 200, 300, 400), not aliases
- Token expired? Repeat `env activate`

## From your instructors

- PA host, tenant ID and PA API key
- Five COS values: endpoint, region, bucket, access key, secret
- PA purchase ID
- Agent YAML and `publish_report_pdf.py`

## You should already have

- Python 3.11–3.13 and `openssl`
- ADK installed and wxO environment active
- PA admin access (needed for the embed step)

**Ready.** Open the guide, run the Part 1 check, then start Part 2.

Q&A in the last 5 minutes